

Overall Architecture of a Multicore Out-of-Order Pipelined RISC-V CPU

1 Target Design

This project designs a multicore pipelined RISC-V CPU based on the RISC-V ISA.

The target architecture is:

- ISA: RV32I as the base ISA
- Optional extensions: M, A, Zicsr, Zifencei
- Core count: 2 cores
- Pipeline style: out-of-order execution
- Frontend: branch prediction
- Backend: register renaming, reservation stations, reorder buffer
- Memory system: shared memory with memory arbiter
- Memory ordering: RISC-V FENCE support

The reason to start with RV32I is that RV32I is the simplest complete RISC-V integer ISA. It contains the basic instructions needed for arithmetic, control flow, load/store, and system-level instruction handling. RV64I can be added later by extending the datapath from 32 bits to 64 bits.

2 Overall Architecture





